Powers of Ten

Times 10: every digit moves one place to the left.

34 × 10 = 340

thousands	hundreds	tens	ones		tenths	hundredths	thousandths
		3	4	.			
	3	4	0	.			

Each digit is worth 10 times as much, so it moves one place left.
The empty ones place gets a 0.

4.6 ÷ 10 = 0.46

			4	. 6		
			0	. 4	6	

Divide by 10: one place right.
Divide by 100: two places right.

Powers of ten: how many 10s you multiply

$10^2 = 10 \times 10 = 100$

$10^3 = 10 \times 10 \times 10 = 1{,}000$

$10^4 = 10{,}000$

The small number says how many 10s, and how many zeros.

Your turn!

$5.2 \times 100 =$ 520

For teachers

Goal: multiply and divide by 10, 100 and 1,000 by moving digits in the place-value chart, and powers of ten (5.NBT.1–2). Say “The digits move; the point stays.” Never “add a zero”.

Powers of Ten: Practice

Times: digits move left. Divide: digits move right.

Part 1 Move the digits Write the new number in the second row.

1)

th	h	t	o	.	tth	hth	thth
		2	7	.			
	2	7	0	.			

$27 \times 10 =$ 270

2)

th	h	t	o	.	tth	hth	thth
			6	.	4		
			0	.	6	4	

$6.4 \div 10 =$ 0.64

3)

th	h	t	o	.	tth	hth	thth
			3	.	1	5	
	3	1	5	.			

$3.15 \times 100 =$ 315

4)

th	h	t	o	.	tth	hth	thth
	8	2	0	.			
			8	.	2		

$820 \div 100 =$ 8.2

5)

th	h	t	o	.	tth	hth	thth
			0	.	9		
			9	.			

$0.9 \times 10 =$ 9

Powers of Ten: Practice (continued)

Part 1 Move the digits (continued)

6)

th	h	t	o		tth	hth	thth
	4	0	5	.			
		4	0	.	5		

405 ÷ 10 = 40.5

Part 2 Times 10, 100 and 1,000 No help this time.

7) 47 × 100 = 4,700

8) 3.6 × 10 = 36

9) 0.25 × 100 = 25

10) 8 × 1,000 = 8,000

11) 5.04 × 100 = 504

12) 62 × 10 = 620

13) 0.7 × 1,000 = 700

14) 19 × 1,000 = 19,000

15) 2.38 × 10 = 23.8

Powers of Ten: Practice (continued)

Part 2 Times 10, 100 and 1,000 (continued)

16) $0.06 \times 100 = 6$

17) $715 \times 10 = 7{,}150$

18) $4.5 \times 1{,}000 = 4{,}500$

19) $0.81 \times 10 = 8.1$

20) $33 \times 100 = 3{,}300$

21) $1.9 \times 100 = 190$

Part 3 Divide by 10, 100 and 1,000 No help this time.

22) $4{,}500 \div 100 = 45$

23) $36 \div 10 = 3.6$

24) $7 \div 100 = 0.07$

25) $8{,}000 \div 1{,}000 = 8$

26) $52 \div 100 = 0.52$

27) $9.4 \div 10 = 0.94$

Powers of Ten: Practice (continued)

Part 3 Divide by 10, 100 and 1,000 (continued)

28) $630 \div 10 =$ 63

29) $1{,}250 \div 1{,}000 =$ 1.25

30) $3 \div 10 =$ 0.3

31) $74 \div 100 =$ 0.74

32) $580 \div 100 =$ 5.8

33) $2.6 \div 10 =$ 0.26

34) $9{,}000 \div 100 =$ 90

35) $45 \div 1{,}000 =$ 0.045

36) $0.8 \div 10 =$ 0.08

Part 4 Powers of ten No help this time.

37) $10^2 =$ 100

38) $10^4 =$ 10,000

Powers of Ten: Practice (continued)

Part 4 Powers of ten (continued)

39) $10^1 = \boxed{10}$

40) $10^5 = \boxed{100,000}$

41) $3 \times 10^2 = \boxed{300}$

42) $7 \times 10^3 = \boxed{7,000}$

43) $2.5 \times 10^2 = \boxed{250}$

44) $10^3 = \boxed{1,000}$

45) $6 \times 10^4 = \boxed{60,000}$

46) $4.2 \times 10^1 = \boxed{42}$

Part 5 Times what? No help this time. Count how many places the digits moved.

47) $52 \times \boxed{100} = 5,200$

48) $3.7 \times \boxed{10} = 37$

49) $0.4 \times \boxed{1,000} = 400$

Powers of Ten: Practice (continued)

Part 5 Times what? (continued)

50) $16 \times \boxed{1{,}000} = 16{,}000$

51) $2.09 \times \boxed{100} = 209$

52) $81 \times \boxed{10} = 810$

53) $0.35 \times \boxed{100} = 35$

54) $7 \times \boxed{100} = 700$

Part 6 Times 10, 100 and 1,000

55) $682 \times 1{,}000 = \boxed{682{,}000}$

56) $607 \times 100 = \boxed{60{,}700}$

57) $237 \times 10 = \boxed{2{,}370}$

58) $310 \times 100 = \boxed{31{,}000}$

59) $55.8 \times 100 = \boxed{5{,}580}$

60) $34.6 \times 100 = \boxed{3{,}460}$

Powers of Ten: Practice (continued)

Part 6 Times 10, 100 and 1,000 (continued)

61) $5.45 \times 10 =$ 54.5

62) $2.15 \times 1{,}000 =$ 2,150

63) $4.16 \times 1{,}000 =$ 4,160

64) $52 \times 10 =$ 520

65) $539 \times 1{,}000 =$ 539,000

66) $4.96 \times 1{,}000 =$ 4,960

67) $17.2 \times 10 =$ 172

68) $3.1 \times 1{,}000 =$ 3,100

69) $985 \times 100 =$ 98,500

70) $264 \times 1{,}000 =$ 264,000

71) $385 \times 10 =$ 3,850

72) $8.32 \times 100 =$ 832

Powers of Ten: Practice (continued)

Part 6 Times 10, 100 and 1,000 (continued)

73) 405 × 10 = 4,050

74) 3.16 × 100 = 316

75) 1.35 × 1,000 = 1,350

76) 349 × 1,000 = 349,000

77) 3.32 × 10 = 33.2

78) 58 × 100 = 5,800

Part 7 Divide by 10, 100 and 1,000

79) 6,898 ÷ 100 = 68.98

80) 245 ÷ 100 = 2.45

81) 3,505 ÷ 1,000 = 3.505

82) 7,911 ÷ 1,000 = 7.911

83) 5,342 ÷ 10 = 534.2

84) 3,504 ÷ 100 = 35.04

Powers of Ten: Practice (continued)

Part 7 Divide by 10, 100 and 1,000 (continued)

85) 7,984 ÷ 1,000 = 7.984

86) 9,978 ÷ 10 = 997.8

87) 6,099 ÷ 10 = 609.9

88) 7,192 ÷ 1,000 = 7.192

89) 6,087 ÷ 1,000 = 6.087

90) 5,447 ÷ 1,000 = 5.447

91) 3,345 ÷ 10 = 334.5

92) 5,160 ÷ 10 = 516

93) 1,694 ÷ 100 = 16.94

94) 9,459 ÷ 1,000 = 9.459

95) 3,635 ÷ 10 = 363.5

96) 9,926 ÷ 1,000 = 9.926

Powers of Ten: Practice (continued)

Part 7 Divide by 10, 100 and 1,000 (continued)

97) 1,450 ÷ 100 = 14.5

98) 9,342 ÷ 10 = 934.2

99) 3,067 ÷ 100 = 30.67

100) 9,635 ÷ 1,000 = 9.635

101) 5,628 ÷ 10 = 562.8

102) 1,958 ÷ 100 = 19.58

Part 8 Remember? Problems from earlier packets.

103) 0.62 > 0.6

104) $4 \times \frac{7}{10} = \frac{28}{10}$

105) area: 18 × 7 = 126 sq ft

106) 48 × 25 = 1,200

107) 6 L = 6,000 mL

108) 40 + 140 = 180 degrees

Powers of Ten: Practice (continued)

Part 9 Word problems Think about how the digits move.

109) A box holds 24 pencils. How many pencils are in 100 boxes?

Number sentence 24 × 100 = 2,400

Answer 2,400 pencils

Draw or show your work

110) A stack of 10 coins is 1.8 cm tall. How tall is one coin?

Number sentence 1.8 ÷ 10 = 0.18

Answer 0.18 cm

Draw or show your work

111) A sticker costs $0.35. How much do 10 stickers cost?

Number sentence 0.35 × 10 = 3.5

Answer $3.50

Draw or show your work

112) A farm has 4,500 apples to pack in 100 equal bags. How many apples go in each bag?

Number sentence 4,500 ÷ 100 = 45

Answer 45 apples

Draw or show your work

113) Challenge: A paper is 0.1 mm thick. How thick is a stack of 1,000 papers, in cm? (10 mm = 1 cm)

Number sentence 0.1 × 1,000 = 100 mm = 10 cm

Answer 10 cm

Draw or show your work

Check-In: Powers of Ten

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Move the digits

1)

th	h	t	o	tth	hth	thth
			5 .	2		
			0 .	0	5	2

$5.2 \div 100 =$ 0.052

Part 2 Times 10, 100 and 1,000

2) $13 \times 10 =$ 130

3) $9.1 \times 1{,}000 =$ 9,100

4) $740 \times 10 =$ 7,400

Part 3 Divide by 10, 100 and 1,000

5) $56 \div 10 =$ 5.6

6) $4{,}100 \div 100 =$ 41

7) $4{,}100 \div 10 =$ 410

Part 4 Powers of ten

8) $9 \times 10^3 =$ 9,000

9) $1.5 \times 10^3 =$ 1,500